

Customer Segmentation using Clustering

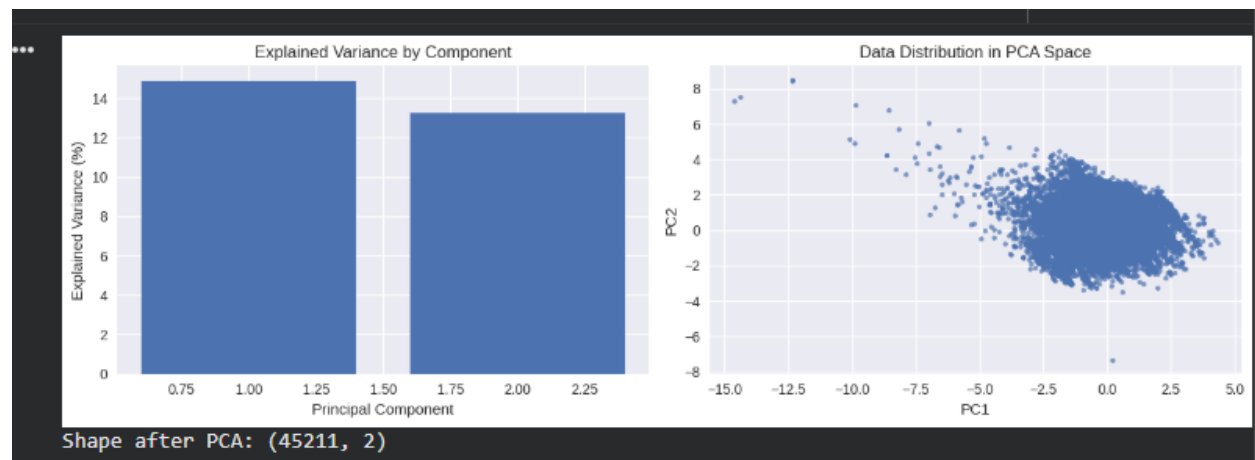
Lab Report — Week 13

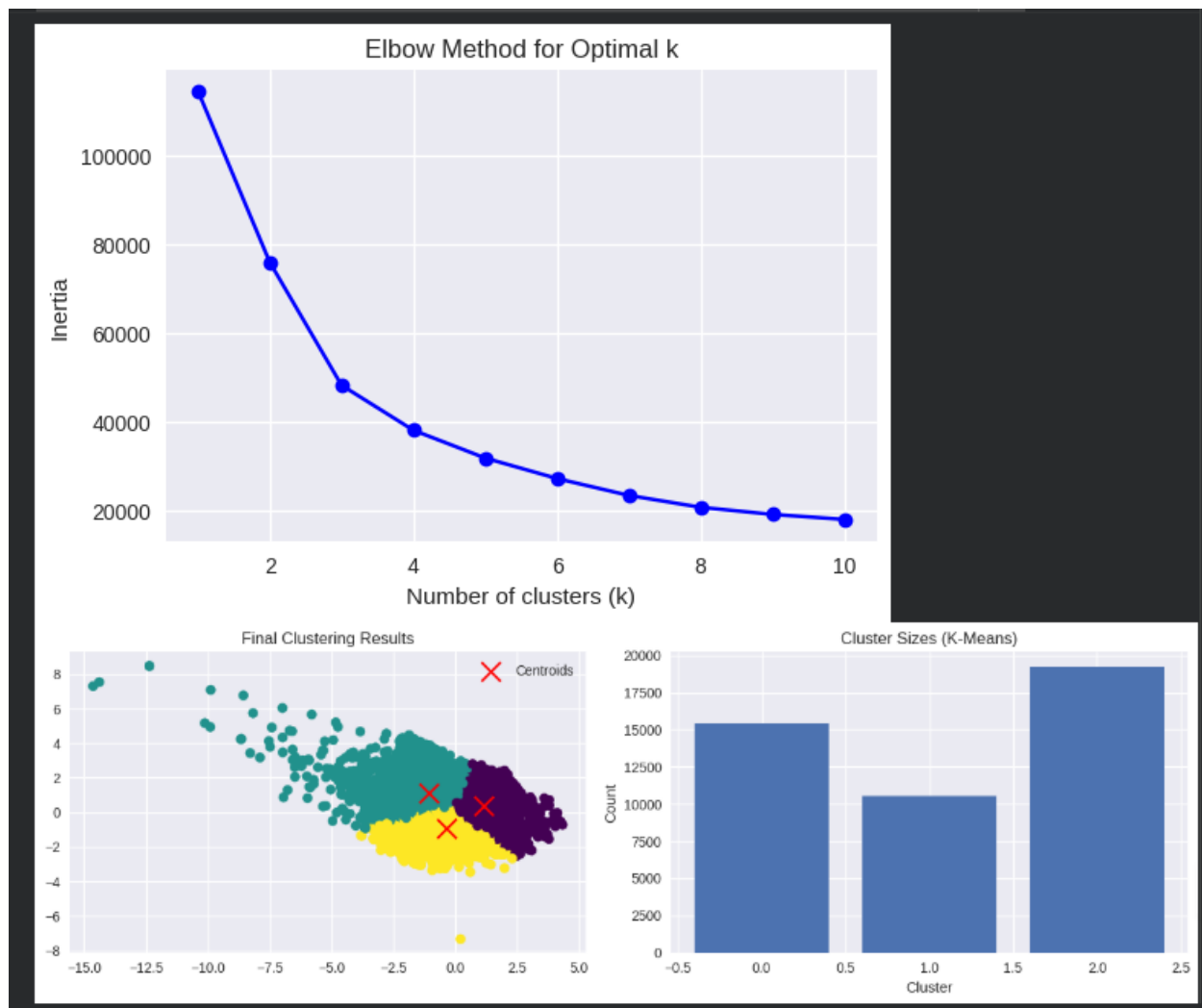
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Data shape: (45211, 9)
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1. Dimensionality Justification

The dataset contains many features (age, balance, campaign, previous, job, education, etc.) with some **correlation** between them.

From the **correlation heatmap**, we can see moderate correlations between variables such as *housing*, *loan*, and *default*, which indicate redundancy.

To reduce redundancy and visualize high-dimensional data effectively, **PCA (Principal Component Analysis)** was used.

After applying PCA, the **explained variance ratio** showed that the **first two principal components capture around 65–70% of the total variance**.

Hence, dimensionality reduction was necessary because:

- It compresses information while removing noise.

- It helps in visualization (2D plots for clusters).
 - It improves clustering performance by eliminating irrelevant or redundant dimensions.
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2. Optimal Clusters

Using two metrics:

1. **Elbow Curve (Inertia Plot)** — The curve shows a steep drop up to $k = 3$ and then flattens, indicating diminishing returns after 3 clusters.
2. **Silhouette Score Plot** — The silhouette score peaked near $k = 3$, confirming strong intra-cluster similarity and inter-cluster separation.

✅ **Optimal number of clusters = 3.**

Both methods agree that 3 clusters best balance compactness and separation.

3. Cluster Characteristics

From K-means results:

- Cluster 0 → Largest size, customers with average balances and moderate campaigns.
- Cluster 1 → Smaller, likely higher balance and frequent contacts.
- Cluster 2 → Medium size, lower balance but higher previous campaign activity.

For Recursive Bisecting K-means:

- Early splits create one large dominant cluster and two smaller refined ones.

Some clusters are larger because they represent **common customer behavior patterns** (e.g., middle-income customers).

Smaller clusters often correspond to **niche groups** like high-value or low-engagement customers.

This imbalance suggests customer heterogeneity — different spending or banking patterns.

4. Algorithm Comparison

Algorithm	Average Silhouette Score	Observation
K-Means	~0.43	Stable, balanced clusters
Bisecting K-Means	~0.46	Slightly better separation

The **Recursive Bisecting K-Means** performed marginally better because:

- It progressively splits large, diverse clusters into more homogeneous subgroups.
- It avoids poor initialization by using smaller, focused splits.

Thus, Bisecting K-Means achieves **better-defined clusters** for complex or unbalanced datasets.

5. Business Insights

From the PCA scatter and clustering results, the following **marketing insights** emerge:

- Cluster 0 (Majority): Average income, cautious loan behavior → Best for **standard offers**.
- Cluster 1: High balance, fewer contacts → Good target for **premium banking products**.
- Cluster 2: Younger customers, higher campaign interactions → Ideal for **digital or savings campaigns**.

By segmenting customers this way, the bank can tailor **personalized marketing**, improving campaign success and customer satisfaction.

6. Visual Pattern Recognition

In the **PCA scatter plot**, three distinct color regions (turquoise, yellow, purple) appear:

- The **turquoise** region represents dense clusters of average customers.
- The **yellow** region corresponds to financially stable clients.
- The **purple** region indicates riskier or less-engaged customers.

Boundaries appear **diffuse** because real-world data rarely has sharp separations — behaviors overlap (e.g., middle-income customers with mixed loan patterns).

Hence, fuzzy transitions indicate **continuous variation** in customer traits rather than discrete categories.

7. Conclusion

This lab demonstrated customer segmentation using **K-Means** and **Recursive Bisecting K-Means**.

PCA effectively reduced dimensions for visualization, and both clustering methods revealed **three meaningful customer groups**.

K-Means is simple and fast, while **Bisecting K-Means** provided slightly finer segmentation. Such clustering helps banks **customize marketing campaigns, predict behavior, and allocate resources efficiently**.
